Dimensional Modeling for Banking Transaction Analytics and Reporting

1. Introduction

This project focuses on designing a dimensional data model for banking transaction analytics. The objective is to transform raw transactional data into a structured warehouse model that supports efficient reporting and decision-making across customer behavior, product performance, branch operations, and time-based analysis. Given the analytical requirements, the solution is designed for OLAP (Online Analytical Processing) workloads, prioritizing historical analysis and aggregation, rather than OLTP (Online Transaction Processing) systems, which are optimized for real-time transaction handling.

2. Data Understanding & Modeling Approach

The dataset consists of transaction-level records containing customer, product, branch, channel, and time-related attributes. Each record represents a single transaction, establishing the grain of the fact table as ***one row per transaction event***.

A dimensional modeling approach was adopted to separate measurable transactional data from descriptive business context. Transactional measures such as amount and balance_after transaction form the fact layer, while descriptive attributes such as customer, product, branch, and channel are structured into dimension tables.

To ensure analytical consistency and independence from operational systems, ***surrogate keys (SKs)*** were introduced across all dimension tables as primary keys. The fact table references these keys as foreign keys, enabling efficient joins and stable relationships. The transaction identifier (*txn_id*) is retained within the fact table as a ***degenerate dimension***, providing traceability without introducing an additional dimension table.

3. Schema Design

The model follows a ***star schema architecture***, with a central fact table connected directly to multiple dimension tables through surrogate keys. This structure simplifies query patterns, reduces join complexity, and improves performance in analytical tools such as Power BI.

The fact table captures transactional events at the defined grain, while dimension tables provide the contextual attributes required for slicing and aggregating data across business perspectives. By relying on surrogate keys instead of natural business keys, the model ensures stability and supports scalable analytical workloads.

4. **Handling Historical Changes (SCD Strategy)**

Certain business attributes are time-variant, particularly customer-related fields such as tier classification. To preserve historical accuracy, a ***Slowly Changing Dimension Type 2 (SCD Type 2)*** approach is applied to the customer dimension, allowing multiple versions of a record to exist over time.

Other dimensions, such as branch and product, are managed using ***SCD Type 1***, where updates overwrite existing values. This approach is appropriate where historical tracking is not required and only the most current state is relevant for analysis.

5. **Data Integration & Processing Logic**

The data pipeline follows an ***ETL (Extract, Transform, Load)*** approach, allowing data to be transformed into a structured dimensional format before loading into the warehouse. This ensures consistency, data quality, and alignment with the analytical model.

Historical data spanning approximately 18 months is first loaded into dimension tables to generate surrogate keys, followed by the fact table which references these keys. For ongoing operations, incremental loads are handled using transaction timestamps to capture new records efficiently.

Data quality is enforced through validation checks including duplicate detection, null validation on key fields, referential integrity between fact and dimension tables, and validation of transaction values.

6. **Performance & Scalability Considerations**

To optimize query performance, indexing is applied on foreign keys within the fact table to improve join efficiency. Additionally, partitioning the fact table by date supports faster execution of time-based queries, which are common in analytical workloads.

To further enhance performance for frequently accessed reports, pre-aggregated tables such as monthly transaction totals by branch can be introduced. These reduces the need to repeatedly scan large volumes of transactional data.

7. **Conclusion**

The resulting dimensional model provides a scalable and efficient foundation for banking transaction analytics. By leveraging a star schema design, surrogate keys, and appropriate SCD strategies, the model ensures both performance and historical accuracy.

It supports key analytical use cases including customer segmentation, product and channel performance analysis, branch-level reporting, churn detection through transaction patterns, and time-based trend analysis such as MoM, QoQ, and YoY comparisons. The final design is robust, scalable, and aligned with modern business intelligence requirements.